Reading a Penalty from Broadcast Video

Computer vision analysis of World Cup penalties

IND5003 Vision II | Group 19
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An analysis of what broadcast footage can reveal about a penalty kick, how the methods improved, and where independent performance remains limited.

Repository: https://github.com/Gabric-cau/ind5003-vision-ii

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Results on the 39 development kicks and on eight independent test kicks are reported separately. Coverage indicates that an output was produced; it does not establish accuracy. The four camera-angle timing outputs remain unresolved (Appendix A).

Abstract

We analyse 39 penalty kicks from four 2018 FIFA World Cup shoot-outs using television broadcasts. The pipeline extracts contact time, kicking foot, approach steps, goalkeeper movement and direction, ball trajectory and outcome when the footage supplies sufficient evidence. Missing or ambiguous observations are recorded as unknown with a reason.

On the development set, contact time, ball trajectory and fused outcome are available for 33 to 35 of 39 kicks. The fused outcome agrees with the official record in all 35 answered cases. The rule-based kicking-foot and goalkeeper-direction outputs agree with a single human reviewer in 12 of 14 and 19 of 23 cases, respectively. Temporal repairs and paired pose checks provide modest gains; small supervised heads trained on frozen features produce the largest improvements. Kicking-foot macro-F1 rises from 0.50 to 0.72, and matched human landings rise from 13 to 41 to 125 of 156. Controlled data augmentation provides no further gain.

The limits remain substantial. Strict dive height, final ball position and the requested camera-angle timing outputs are unresolved. On eight unseen penalties from the 2022 World Cup final, all runs complete, but kicking-foot strict accuracy is 3 of 8 and the outcome and height rules abstain throughout. The findings therefore support reproducibility and development improvements, while showing that reliable generalisation requires more diverse footage and stronger reference labels.

1 Literature review of similar approaches

1.1 Penalty kicks as a research subject

Penalty kicks are short, repeated contests with a clear outcome, making them useful for studying decision-making. Chiappori et al. [1] and Palacios-Huerta [2] found choices of side close to mixed-strategy equilibrium and a natural side associated with kicking foot. Bar-Eli et al. [3] analysed 286 penalties and found that goalkeepers usually dived even when staying central could be effective. They interpreted this preference as action bias.

These studies relied on manual video coding. Automating the same variables, including kicking foot, goalkeeper direction and outcome, could support larger samples. Our project examines whether broadcast footage supplies enough evidence for that automation.

1.2 Computer vision for sports video

SoccerNet [4] supports temporal action spotting in football broadcasts. Our task requires finer event detail: the contact instant, striking foot, approach landings and goalkeeper response within one penalty.

YOLO-family detectors [5, 6] locate people and the ball in each frame; temporal association is still needed to maintain identity.

For human pose estimation, the COCO benchmark [7] provides 17 keypoints ending at the ankle. BlazePose [8], implemented through MediaPipe [9], estimates 33 landmarks on a person crop, including heel and toe points useful for landings. Small players, occlusion and motion blur limit both approaches. SAM 2 [10] follows a prompted object mask; we tested it for goalkeeper outlines.

For camera geometry, Shi–Tomasi features [11], Lucas–Kanade tracking [12] and RANSAC [13] estimate camera motion. Hough pitch lines [14] and goal references support a homography to the ground plane when sufficient geometry is visible [15]. Regularised logistic regression on frozen ResNet features [16] and gradient-boosted trees [17] provide compact models for small labelled samples.

1.3 Research gap and design principle

Broadcast penalties combine camera cuts, replays, slow motion and occlusion at contact. We therefore require each module to meet an evidence threshold before reporting a value; otherwise it returns unknown with a reason. This separates missing observations from model errors. Section 9 assesses the analysis against the principles of McGowan et al. [18], including scepticism, clarity and reproducibility.

2 Data overview and preparation

2.1 Main dataset

The main dataset contains 39 penalty kicks from four 2018 World Cup shoot-outs supplied by the course, shown in Table 1.

Table 1. Main dataset

Match	Penalties
Colombia vs England	10
Croatia vs Denmark	10
Spain vs Russia	9
Russia vs Croatia	10
Total	39

Each penalty is cut from the original recording with FFmpeg and decoded with OpenCV. For every clip we record the original frame numbers, the frame rate, the clip offset and a SHA-256 hash of the source file. Every output can therefore be traced back to the exact frames it came from, and a rerun on a changed file is detected rather than silently mixed with older results.

2.2 Reference labels

We used four kinds of reference information. The official outcome of all 39 kicks comes from FIFA match records, with 26 goals and 13 misses or saves. One group member labelled all 39 kicks by hand, recording landing frames (228 observations, of which 223 fall inside the automatic review windows), the kicking foot (36 right-footed and 3 left-footed kicks) and four qualitative shot and dive labels. A second independent review was not completed, so these are single-reviewer labels and we cannot measure agreement between reviewers. A further set of 24 label updates refines existing events; it does not add new penalties. Lastly, six short clips with 835 annotated goalkeeper bounding boxes were used to train a dedicated goalkeeper detector.

2.3 Additional data collected to strengthen the findings

Because 39 kicks from four matches is a small sample, we looked for additional footage that could either support training or provide an independent check. All additional sources were kept separate from the main dataset and were given a fixed role before any results were computed.

The most important addition is a frozen test set of eight penalties from the 2022 World Cup final between Argentina and France. These clips were not decoded, inspected or used for any training or setting choice until the end of the project, when the final pipeline was run on them once (Section 6.3).

A teammate also assembled a release of club-match penalties with human labels, from which we used eight events from the 2022 German Cup for training and seven from the 2025 Coppa Italia for validation of experimental outcome and ball-category models. One event in this release was excluded because its recorded human outcome and its external result conflicted. For the ball detector we added 320 labelled frames from Bayer Leverkusen footage for training and 88 from Arsenal footage as a comparison group.

A separate set of ten core penalty clips came with human-viewed contact times and outcomes, which we used to train and validate a contact-time model. To test whether image quality was the main obstacle, we also obtained 1080p, 50 frames-per-second recordings of three kicks (Falcao, Kane and Cuadrado). Finally, we reviewed the public SoccerNet Tracking, Calibration and Game State Reconstruction datasets. We decided not to train on them because none of them contains kicking-foot, landing or dive labels for penalties, but they informed our choice of tracking and calibration methods.

2.4 Partitions and leakage control

All frames and camera views of a penalty stay in the same partition, so training cannot use one view of a kick that also appears in the test partition. The learned kicking-foot and landing models on the 39-kick dataset are evaluated by leave-one-match-out validation: each model is trained on three matches and tested on the fourth. Scaling and dimensionality reduction are fitted inside each training fold only. The separate contact-time and external outcome experiments use the datasets and evaluations described in Sections 2.3 and 5.4–5.5. Because the 39-kick dataset was inspected throughout development, its results are development results. The eight penalties from the 2022 final provide the fully independent check.

Original match videos, model weights and personal-account data are not committed to the public repository. The reproduction package documents where these files must be placed (Section 8).

3 Exploratory data analysis

3.1 Output coverage and observability

The first finding of the exploratory analysis is that the requested features differ greatly in how often they can be observed. Table 2 shows how many of the 39 kicks received an output from the rule-based pipeline for each feature, and Figure 1 shows the same coverage as a chart.

Table 2. Output coverage of the rule-based pipeline on the 39-kick development set

Feature	Kicks with an output	Notes
Contact time	33/39 (84.6%)	All 10 kicks in Colombia vs England; the contact stage did not run for 6 kicks in the other matches
Kicking foot	14/39 (35.9%)	13 in the main table plus 1 from the supplement; 10 right, 4 left
Approach steps (ankle-based count)	32/39 (82.1%)	Older method counting ankle-height extremes; not validated as landings
Approach steps (landing candidates)	12/39	22 landing candidates in total after the October 18 repair
Goalkeeper movement before contact	29/39 (74.4%)	Measured in image space
Goalkeeper dive direction	23/39 (59.0%)	Screen left, screen right or centre
Ball trajectory	33/39 (84.6%)	Tracked observations with recorded gaps
Ground projection	32/39	226 pre-contact ground points; no continuous top-down path
Outcome (fused)	35/39 (89.7%)	Scoreboard and ball-trajectory evidence combined
Strict dive height	0/39	Required body, ground and contact references not visible
Strict final ball position	0/39	14 occluded, 8 cut away by the camera, 17 undetermined
Slow-motion classification	0 of 320 camera intervals	Cross-shot alignment not established

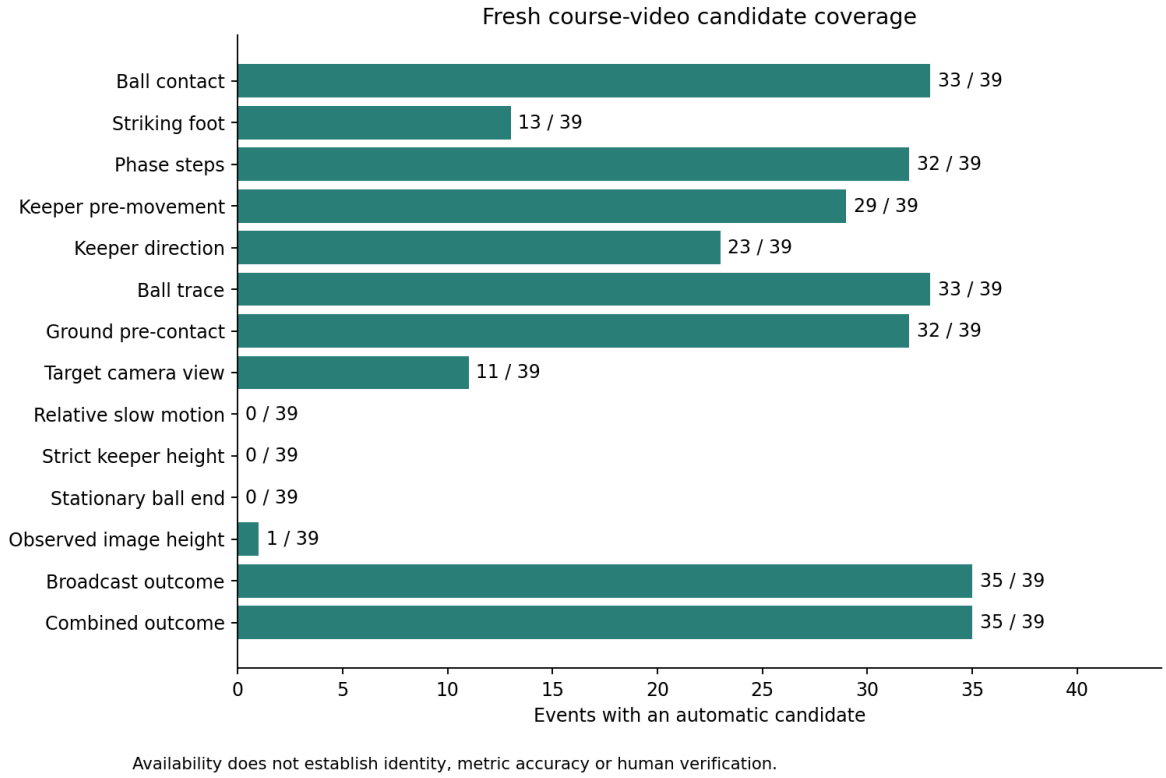


Figure 1. Output coverage per feature for the 39 development kicks. Coverage means that an output was produced, not that it was correct.

The features with the highest coverage depend on large, visible objects or on the broadcast overlay, namely contact, the ball track and the scoreboard. The lowest coverage is for features that need fine body detail at the moment of greatest occlusion, namely the foot, landings and dive height, or a view of the goal after the ball arrives, namely the final position.

3.2 Penalty characteristics

Among the features that could be extracted, the data show several patterns. All counts are small and come from four matches, so they describe this sample rather than general behaviour.

According to FIFA records, 26 of 39 kicks (66.7%) were scored. Of the 14 kicks with a foot output, 10 were right-footed and 4 left-footed. Of the 23 kicks with a dive-direction output, the goalkeeper dived to screen-left in 12, to screen-right in 9 and stayed central in 2. Diving to one side in 21 of 23 cases is consistent with the action bias reported by Bar-Eli et al. [3], and both central cases were goals. Directions are recorded in screen coordinates, and because camera positions differ between matches, screen-left does not always mean the goalkeeper's own left. Of the 29 kicks with a goalkeeper-movement output, 27 show measurable movement before the ball is struck and 2 show the goalkeeper essentially still. Both still cases were saves or misses, but two cases do not support any conclusion, and because movement is measured in image space, small steps cannot yet be fully separated from tracking noise.

The visible approach lasted a median of 1.8 seconds (n = 32, range 0.7 to 4.2 seconds). The ankle-based step count had a median of 5 (range 0 to 15), and it was similar for scored and unscored kicks (Figure 2). The extreme values of 13 and 15 come from long run-ups or tracking errors.



Figure 2. Ankle-based step candidates by outcome (left) and their sensitivity to the detection threshold (right). The outlier of 15 steps is Rashford's kick, for which the later stance-based method found only 3 visible stance changes, which suggests that the ankle-based method over-counts.

3.3 Agreement with human labels

Table 3 compares the outputs of the rule-based pipeline with the single human reviewer. Only kicks where the system produced an output are compared, and unknowns are never counted as correct.

Table 3. Agreement of the rule-based pipeline with single-reviewer human labels (development set)

Feature	System outputs	Agree with human	Notes
Kicking foot	14	12 (85.7%)	Disagreements: Golovin, Brozovic
Goalkeeper dive direction	23	19 (82.6%)	4 disagreements retained
Landings, original rules	22 candidates in 27 comparable windows	13 of 156 human landings	Precision 59.1%, recall 8.3%, count MAE 4.963
Landings, with heel/toe support	65 candidates	41 of 156 human landings	Precision 63.1%, recall 26.3%, count MAE 3.370

The foot and direction results are encouraging where an output exists, but they cover only about a third and about three fifths of the kicks respectively. Landing detection was the clearest weakness of the rule-based pipeline.

3.4 Outcome evidence

The scoreboard method produced an outcome for 35 kicks and agreed with FIFA records in all 35 (Figure 3). The ball-trajectory method produced an outcome for only 10 kicks and agreed in 9. The two sources cover slightly different kicks. For Koke, the ball trajectory and the scoreboard conflict, so the fused result stays unknown. For Rakitic, the scoreboard is missing and the ball-trajectory result is used. These agreement rates were measured on development data and are not independent test accuracy.

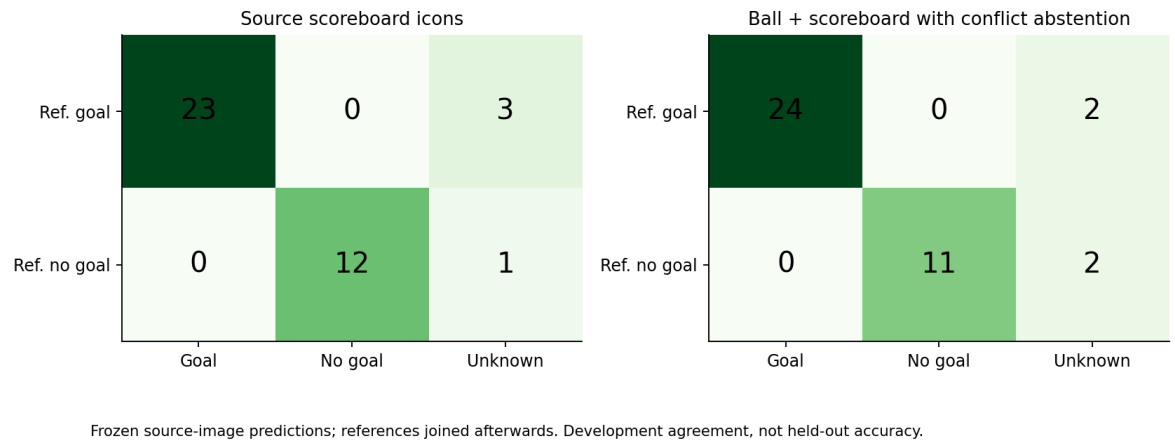


Figure 3. Outcome from the scoreboard icons alone (left) and fused with the ball trajectory, with abstention on conflict (right), compared with FIFA records. No kick was assigned the wrong outcome.

3.5 How the exploratory analysis directed the next steps

The exploratory results shaped the rest of the project in three ways. First, the zero coverage for dive height and final ball position, together with the reasons recorded for each unknown, showed that these gaps are caused by what the camera shows. We therefore stopped trying to tune models for them and framed this as a research question about observability. Second, the large gap between the high-coverage features and the foot and landing features showed where effort would pay off. We concentrated on the kicking-foot and landing modules, which became the subject of most of our iterations (Section 5). Third, Table 3 showed that landing recall of 8 to 26 percent could not be fixed by adjusting thresholds alone, which led us to collect human landing labels and to train small supervised models on top of the frozen pose features.

4 Statement of questions of interest

The brief and exploratory analysis lead to three questions about observability, method improvement and generalisation. Each question has an explicit evaluation approach and is answered separately in Section 6.

RQ1. Which of the requested penalty features can be observed in broadcast footage, and which cannot, regardless of model quality? We assess this through per-feature coverage and the recorded reasons for unknown outputs. The distinction matters because an output can be missing either through a model failure or because the footage lacks the required visual evidence. Section 6.1 relates the coverage results to these evidence requirements.

RQ2. Which changes actually improve the outputs: temporal rules, paired pose checks, learning from human labels, or data augmentation? We compare successive methods using the same human-reviewed windows where possible, reporting kicking-foot macro-F1, matched landings, precision and count error. The final augmentation experiments use a comparison rule fixed before evaluation. Section 6.2 identifies which changes helped and which did not.

RQ3. Is the system reproducible, and does its development performance carry over to footage from a match it has never seen? Reproducibility is assessed through recorded inputs, hashes, stage outputs and a separate-machine installation. Generalisation is assessed on the eight frozen penalties from the 2022 World Cup final, reporting coverage and strict accuracy with unknowns counted as wrong. Section 6.3 keeps these two forms of evidence separate.

5 Iteration of the analysis

Each analysis revealed something about the data that suggested what to do next. This section describes the sequence of iterations in the order in which they happened, because the final results only make sense in the light of the failures that led to them.

5.1 Auditing inherited results 8 to 9 October

The project began from an earlier pipeline and its saved results. Our first step was an audit of the reference labels. We found that the step labels used to report an earlier accuracy figure had been generated by an AI assistant and never checked by a person. Agreement between the algorithm and these labels said nothing about accuracy, so we withdrew those figures and marked the labels as unverified. We then inspected the 39 saved ball-tracking images and found that 11 of the tracks were clearly following the goalkeeper or the referee instead of the ball. This led us to add identity checks that require the ball to start at rest on the penalty spot and to move toward the goal, and to mark any kick that fails these checks as unknown.

5.2 Fresh rerun and first model comparisons 12 to 13 October

Because the inherited results could not be trusted, we reran the whole pipeline from the source videos with hash-checked inputs. We compared the nano and medium YOLO11 ball detectors on identical inputs. The medium model produced contact candidates for 29 kicks and the nano model for 26, and the two agreed on contact for 22 kicks (Figure 4). This showed that neither model alone was reliable and led to the rule that both detectors must agree before a contact time is reported.

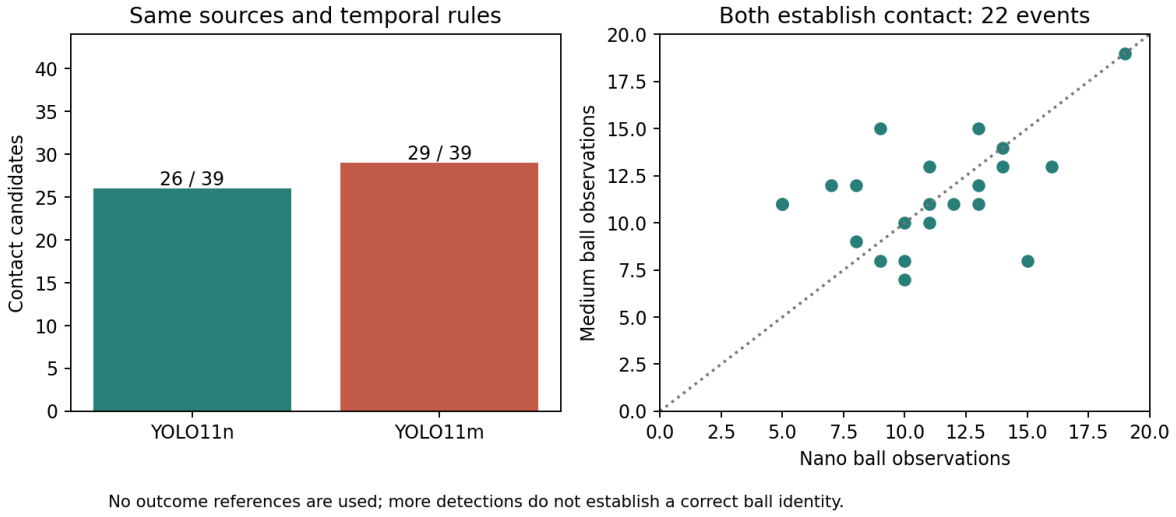


Figure 4. Contact candidates from the nano and medium YOLO11 ball detectors under the same temporal rules (left) and the number of ball observations per kick for the 22 kicks where both establish contact (right).

The kicking-foot module went through three versions in these two days. Version 1 gave a foot for only 7 of 39 kicks. A diagnostic run on its outputs showed two causes. The motion of each frame was divided by that frame's changing detection-box height, so foreshortening during the run-up created apparent motion of both ankles, and the ball-distance window included frames after the ball had already left. Version 2 used one fixed body scale for the whole approach and required the original and mirrored pose passes (Section 7.4) to agree within 0.15 body heights before their points were averaged. Version 3 measured the ankle's approach to the ball's last stationary position instead of to the departing ball, so that observations after contact could no longer act as the target. We also tested whether better image quality would help by running the same

pipeline on 1080p, 50 frames-per-second recordings of three kicks. The higher resolution did not make the foot at contact reliably visible, because the foot is hidden by the body or blurred at that moment rather than simply too small. We therefore did not pursue more footage of the same kind.

5.3 Repairing the landing logic 18 October

When we examined why the two pose passes rarely agreed on landings, we found a bug. Each support period was cut after three low-motion frames, so matching periods that started one frame apart in the original and mirrored passes were never paired. After the repair, the number of landing candidates rose from 12 to 22. We then required the heel or toe point, rather than the ankle, to stay fixed, which is a more direct sign of a planted foot. This raised the number of matched human landings from 13 to 41 out of 156 without lowering precision (Table 3).

5.4 Human labels and learned heads 22 to 26 October

Human labels received on 22 October allowed direct comparison with reviewed observations instead of agreement between automated rules. They also supported small supervised models on frozen pose and detection features. The kicking-foot and landing heads were evaluated by leave-one-match-out validation on the 39-kick dataset. The large pretrained networks remained frozen because the small sample would not support reliable end-to-end training.

The landing classifier scores each foot separately. Its 100-dimensional input combines 19 geometric and motion features at five offsets from 0.12 seconds before to 0.12 seconds after the frame, plus five availability flags. The features describe ankle, heel and toe positions relative to the hips, knee angle, mirror disagreement and camera-stabilised motion. The kicking-foot head uses 57 features summarising the difference between the right and left legs around contact. Training includes negated features with swapped labels to reduce preference for one foot. Because only 3 of 39 kickers are left-footed, an always-right model scores 36 of 39. We therefore select models by macro-F1, which rewards both classes, rather than accuracy.

The learned landing head matched 125 of the 156 human landings in the same 27 windows, with a precision of 0.74 and a count error of 0.89 steps per window, and 123 of its 125 matches were also on the correct foot. The learned foot head gave an answer for 32 kicks and was correct for 30, with a macro-F1 of 0.72 compared with 0.50 for the frozen rule and 0.48 for the always-right control (Figure 5). Both left-footed kicks with usable input were correct. When we later recovered pose inputs for the remaining seven kicks, the same frozen head scored 35 of 39, while retraining on all 39 gave 34 of 39 with a slightly lower macro-F1 of 0.69, so we kept the frozen head.

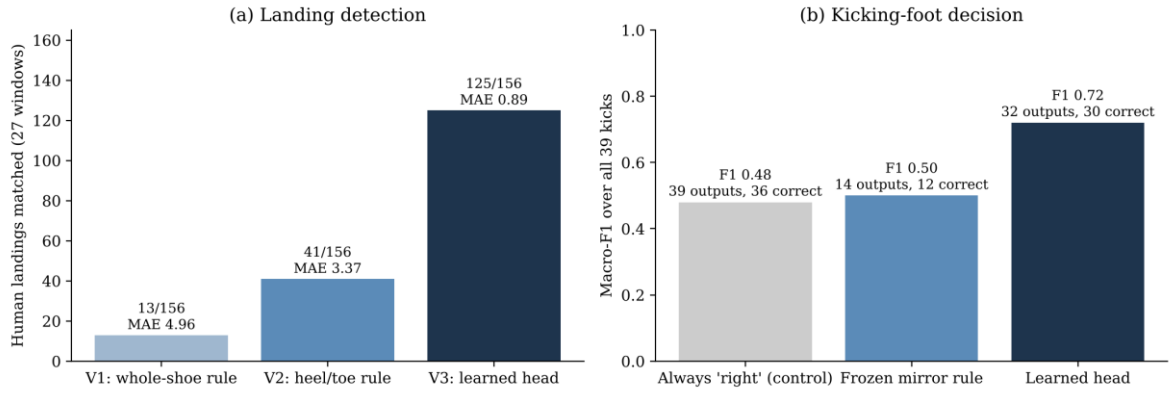


Figure 5. (a) Human landings matched in the same 27 windows by successive versions, with the mean count error per window. (b) Kicking-foot macro-F1 over all 39 kicks, counting unknowns as wrong, for the always-right control, the frozen mirror rule and the learned head.

We applied the same approach to two other outputs. A logistic contact-time head trained on the ten core clips reached a mean absolute error of 0.080 seconds, with all ten predictions within 0.2 seconds of the human timing, compared with 0.113 seconds for the existing rule on the same evidence. A learned goalkeeper-direction head gave more outputs than the rule (29 against 23), but on the 23 kicks covered by both it was correct for 17 against the rule's 19, so it was not allowed to replace the rule.

5.5 Unsuccessful methods retained in the record

Not every attempt worked, and we kept the negative results. A learned outcome model, using frozen ResNet18 features and balanced logistic regression, was trained on the German Cup penalties and scored 7 of 7 on the Coppa Italia validation set. On the ten core clips, however, it predicted "goal" for every kick, which is a classic sign of overfitting to a small and unbalanced sample. Adding the core clips to training reduced the validation score to 3 of 7. Neither version replaced the scoreboard rule. SAM 2 was box-prompted on the goalkeeper to estimate dive height, but the mask often spread onto the ball, and only one kick gave a usable height signal. The rule we followed throughout was that a new method replaces the default only if it beats it on the evaluation fixed in advance.

5.6 Frozen test contact search repair and augmentation 27 to 29 October

When we prepared the frozen test on the 2022 final, we found that the contact extractor only searched the first 12 seconds of the first camera shot. In two of the new clips the kick happens after this point, so no amount of training could have found the contact. We raised the limit to 60 seconds and confirmed that the change left all ten core results unchanged. We then ran the frozen pipeline once on the eight test penalties (Section 6.3). In a final round, we tested data augmentation for the ball and goalkeeper detectors under a comparison rule fixed before training (Section 6.2). It did not help, so we stopped there.

6 Findings for the questions of interest

6.1 RQ1 Which features can be observed

Contact time, ball trajectory, goalkeeper movement before contact and outcome can be extracted for most kicks (29 to 35 of 39). Kicking foot and dive direction can be extracted for a subset, with good agreement

with the human reviewer where an output exists, and the learned foot head extends the foot output to 32 kicks. Figure 6 shows a typical case in which contact can be bracketed to a few frames.



Figure 6. Kane (England vs Colombia) at frames 245, 251 and 257. The ball is at rest until frame 250 and moving by frame 252, so contact is bracketed to frames 250 to 252. The yellow box is the tracked goalkeeper.

Two requested features could not be measured for any kick: strict dive height and the ball's final position. Both need references that broadcast footage rarely provides. Dive height requires the goalkeeper's full body, a visible ground reference and the moment of contact in one shot. The final ball position requires a view of the goal after the ball arrives, but in 22 of the 39 kicks the ball was hidden or the camera had already cut away. An older image-based method produces height candidates for 17 kicks and a last-visible ball frame for 27, but we report these only as provisional observations, not as measurements. The answer to RQ1 is that these gaps are caused mainly by what the camera shows. A better model would not recover a dive that happens off screen.

6.2 RQ2 Which changes improved the outputs

Temporal rules and paired pose checks helped modestly. They raised kicking-foot coverage from 7 to 14 kicks while keeping 12 of 14 in agreement with the human reviewer, and they raised matched landings from 13 to 41. Learning from human labels helped much more. The foot macro-F1 rose from 0.50 to 0.72 and matched landings rose from 41 to 125 of 156 (Figure 5). These are development results under leave-one-match-out validation, not results on an independent test set.

Data augmentation did not help. We ran one final controlled training round for the ball and goalkeeper detectors. In each comparison, the variants shared the same starting checkpoint, data order and number of training updates, and the selection rule was fixed before evaluation.

Table 4. Final augmentation experiments (development groups)

Model	Variant	Group A	Group B	Selected
Ball detector (hits at IoU ≥ 0.5)	Existing starting checkpoint	Arsenal 61/88	Core 65/91	✓
	New control (10 epochs)	58/88	62/91	
	Crop, photometric and affine augmentation	58/88	62/91	
Goalkeeper detector (top-1 IoU ≥ 0.5)	Control (12 epochs)	External match 60/294 (20.4%)	Subject 02 92/98 (93.0%)	✓
	Photometric augmentation	44/294 (15.0%)	92/98 (93.9%)	

Neither new ball variant beat the existing checkpoint, so the default detector was retained. Goalkeeper augmentation reduced external-match performance, so the control was selected. The goalkeeper detector succeeds in about 93 percent of familiar-clip frames but only about 20 percent from another match; it remains an optional diagnostic outside the main pipeline. Augmentation of the same source views did not overcome this transfer gap, supporting the need for more diverse footage and labels.

6.3 RQ3 Reproducibility and generalisation

The saved results are reproducible. Every output can be traced to source hashes and frame timestamps, and the saved-result analysis and review pages can be regenerated locally (Section 8). A portable package was installed from scratch on a separate Windows laptop with an RTX 2060 graphics card and processed a new video through all 14 stages in about 3 minutes 41 seconds.

Generalisation, however, is limited. We ran the frozen pipeline once on the eight penalties from the 2022 final, using no reference labels or manual input during inference.

Table 5. Independent check on eight unseen penalties (2022 World Cup final)

Output	Coverage	Strict accuracy (unknown counted as wrong)	Accuracy where an output exists
Kicking foot	5/8	3/8 (37.5%)	3/5 (60%)
Goal / no goal	0/8	0/8	not defined
Goalkeeper height (ordinal)	0/8	0/8	not defined

All eight independent runs completed, with a median end-to-end time of 287 seconds and a 95th percentile of 468 seconds per video on a shared RTX 4090. Outcome remained unknown without a stable scoreboard change or verified goal-line crossing; height remained unknown without a stable goalkeeper trajectory. Kicking-foot performance fell below development agreement. Thus reproducibility did not establish generalisation to a new match. No model was adjusted using these eight results, preserving their role as an independent test.

7 Overview of new techniques

This section explains the techniques used beyond the course textbook, including pretrained pose estimation, promptable video segmentation, robust camera geometry and small supervised models on frozen features. For each technique, we describe its role, operating principle and limitations. The paired-pose checks and landing repairs in Section 5 show how examining model failures led to revised methods.

7.1 Pose estimation with two model families

We use two pose models, and neither uses a segmentation mask; both regress joint coordinates directly. YOLO11m-pose is a one-stage model. In a single forward pass over the full 1280-pixel frame, each grid cell predicts a person box together with 17 COCO keypoints, and non-maximum suppression removes duplicate detections. This makes it fast enough to run on every frame and gives the kicker and goalkeeper tracks. Because COCO keypoints stop at the ankle, we use MediaPipe's BlazePose (Pose Landmarker Heavy) for the kicker's feet. It is a two-stage, top-down model that first locates the person and then regresses 33 landmarks on a crop, including heel and toe points, which we keep only when their visibility score is at least 0.5. Detections are linked across frames by position, box size and motion continuity, and the kicker is identified as the person who approaches the penalty spot while the goalkeeper is the person on the goal line. If identity becomes ambiguous, for example when players overlap, the downstream outputs for that kick are marked unknown.

7.2 Masks and their roles

Masks are used in four places, always to limit where the system looks and never to decide what an object is. A pitch mask is built from an HSV colour threshold around the dominant grass hue and cleaned with morphological opening and closing. A background mask is the pitch with every person box and the scoreboard area removed, and it is used for camera-motion estimation. Small bright, low-saturation blobs inside the pitch mask provide additional ball candidates when the detector misses the ball. Finally, SAM 2 was tested as a promptable video segmentation model to follow the goalkeeper's outline (Section 5.5).

7.3 Contact time from two ball detectors

Two YOLO11 detectors, nano and medium, look for the ball with a low confidence threshold and tiled inference, because the ball covers only a few pixels. Contact is defined physically: the same ball identity stays still on the penalty spot and then moves toward the goal. The contact time lies between the last stationary frame and the first moving frame, as in Figure 6, and the kicker's ankle must approach the spot within that window. A contact time is reported only when the two detectors agree in at least three shared frames. This removes false contacts caused by a single spurious detection, such as a white boot or a logo.

7.4 Kicking foot with mirrored pose

Pose models often swap the left and right legs, especially when the legs cross during the kick. To detect this, we run the pose model twice, once on the crop and once on a horizontally flipped copy, then flip the second set of points back and swap their left and right labels (Figure 7). A correct estimate puts each leg in the same place in both passes, and frames where the two passes disagree by more than 0.15 body heights are treated as unreliable. Motion is scaled by a fixed body height for the whole approach, so zooming cannot create apparent motion. The striking ankle must approach the ball's resting position and pass through it, while the other leg stays nearly still. The foot is reported only when both passes agree.

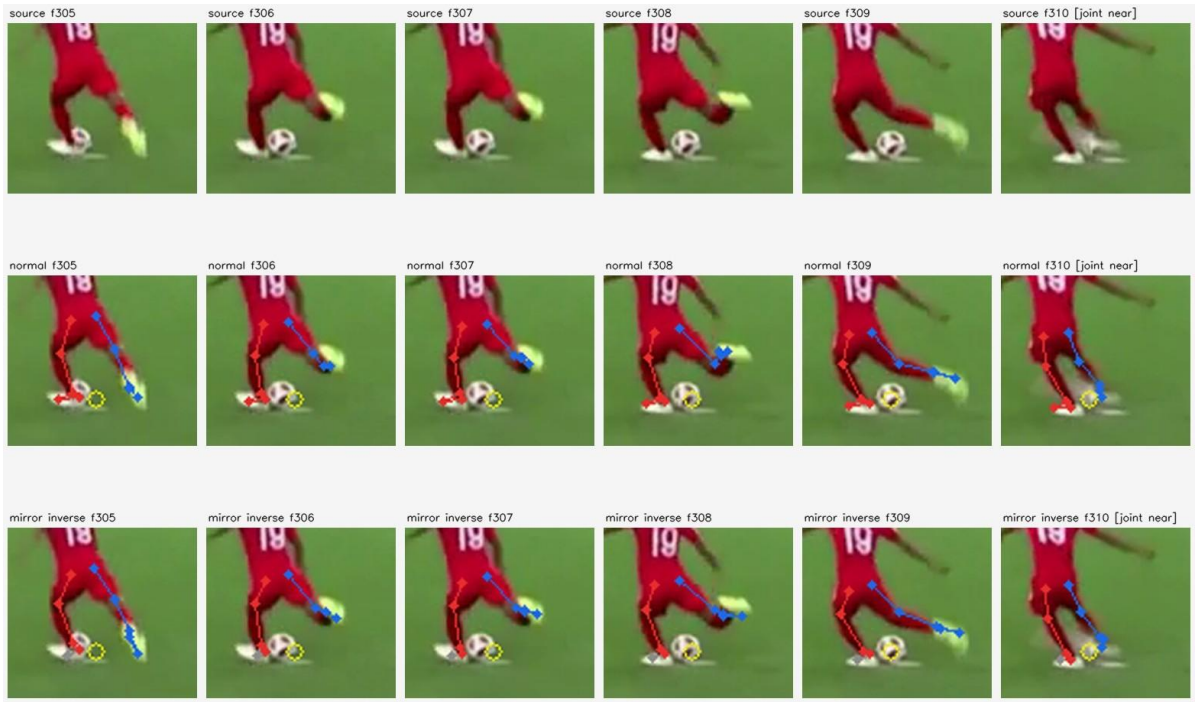


Figure 7. Rashford's kick, frames 305 to 310. Top: source crops. Middle: pose on the original crop. Bottom: pose on the mirrored crop, mapped back. The two passes agree on which leg strikes the ball.

7.5 Landings

A landing is a foot that stops after a swing. After camera-motion correction, heel or toe speed must remain low for at least three frames (0.067 seconds). These points indicate planted support more directly than the ankle (Figure 8). The learned head in Section 5.4 replaces the fixed threshold with frame-level scores.



Figure 8. Sampled frames from Rashford's approach with the original and mirrored poses. The last panel is marked as a support phase on the left foot.

7.6 Camera motion and ground projection

We track up to 350 Shi–Tomasi background corners with pyramidal Lucas–Kanade optical flow and retain forward–backward agreement within 1.5 pixels. RANSAC fits a four-parameter shift, zoom and rotation transform while rejecting moving-player outliers. For ground projection, Hough pitch lines and goal corners provide at least four reference points for a 3 by 3 homography. Of 609 sampled frames, 521 pass the geometry checks, yielding 226 pre-contact ground points (Figure 9). The homography applies to ground-contact feet, not to the head or an airborne ball.

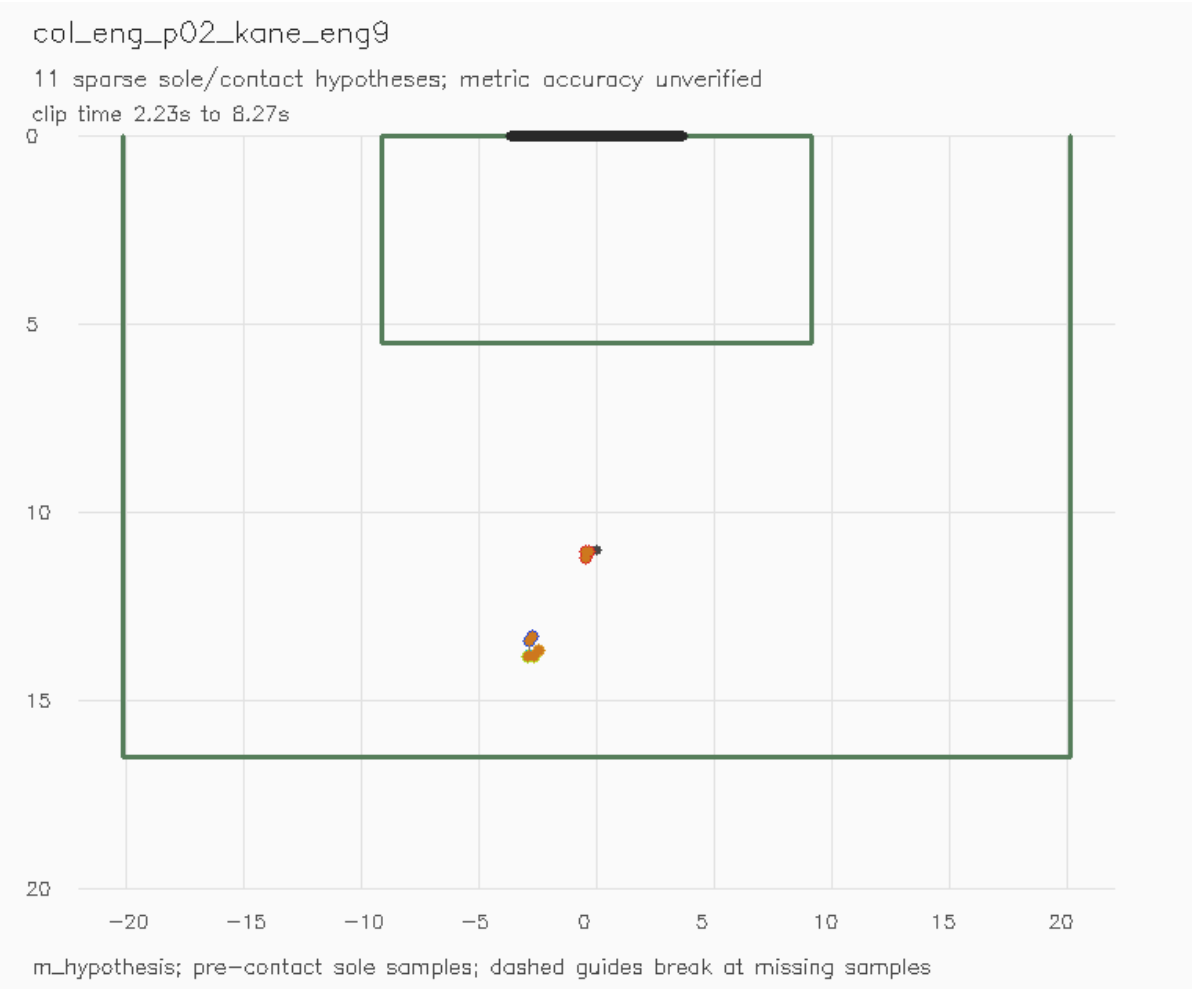


Figure 9. Kane's pre-contact foot positions projected onto the penalty area (11 sparse points). The points are hypotheses; their metric accuracy has not been verified.

7.7 Goalkeeper phase and direction

Pose displacement relative to the goal frame separates the set, pre-contact movement and dive phases. The dominant horizontal displacement after contact gives screen-left, screen-right or centre; these are image coordinates rather than the goalkeeper's own left and right.

7.8 Outcome from the scoreboard

The broadcast overlay contains outcome icons that turn green for a goal and red for a miss. An HSV mask locates the red bar; circularity $4\pi A/P^2$, for area A and perimeter P, identifies the icons. A change must persist across frames. The scoreboard result is fused with the ball-trajectory rule, with unknown returned on disagreement.

7.9 Small supervised heads on frozen features

The landing head uses shallow, regularised histogram gradient-boosted trees; foot and contact heads use standardised logistic regression. The experimental outcome head averages frozen ResNet18 features over four time bins, standardises them, reduces them to seven principal components and fits class-balanced logistic regression. Hyperparameters are not selected on the evaluation folds.

8 Overview of code structure

The code is organised as a Python package with separate folders for reusable modules, experiment scripts, tests, configuration and saved results, as shown in Table 6.

Table 6. Repository layout

Location	Contents
src/	Reusable modules for detection, pose, tracking, masks (field_masks.py), camera motion (motion_stabilization.py), homography (homography.py), goalkeeper, ball and scoreboard (broadcast_scoreboard.py) analysis
scripts/	Experiment entry points, training scripts for the learned heads, single-video inference and audit scripts
tests/	Regression and unit tests
configs/, requirements-*.txt	Configuration and dependencies for each stage
outputs/fresh_course_experiment/	Saved results, EDA figures and review pages
docs/	Experiment protocols, results and reproduction instructions, one document per iteration

The pipeline for a single video runs as a sequence of stages, each of which reads the saved outputs of the previous stage and writes its own outputs together with hashes of its inputs, model and code. A stage refuses to run if any of these hashes does not match, which prevents results from different versions being mixed.

The EDA figures can be regenerated from the saved feature table by installing requirements-analysis.txt and running python -m src.fresh_eda. To process a new video, the portable package in GitHub Release v2026.10.28-portable-v5 provides model weights and an installer for Windows, Linux and macOS on a GPU or CPU; it takes a single penalty clip and writes analysis.json containing the final features and the log of each stage. Each learned head has its own training script, and the final training round records the fixed sampling lists, numbers of optimiser steps, weight hashes and selection rules.

The full test suite recorded 1,543 passed and 4 skipped tests, and the 53 focused tests for the final training round also passed. Original match videos, model weights and ZIP archives are not in the public repository, so to rerun feature extraction from the raw recordings, the course videos and model files must be placed in the documented local paths or in the shared SharePoint folder.

8.1 Review interface

A browser-based review page shows each kick with its source frames, detections and outputs, including the reason for every unknown. A separate blind-review page hides the predictions, so that the human labels could be collected without being influenced by the system’s answers.

9 Limitations quality assessment and conclusion

9.1 Limitations

The most important limitation is that all human labels come from a single reviewer, so the agreement between reviewers, and therefore the noise in our reference, is unknown. There are also no independent

reference coordinates for the ball's final position and no metric measurements of dive height, so these outputs could not be validated even in principle. Slow-motion and camera-type labels were not resolved. The 39-kick dataset was inspected throughout development, and the only independent check covers eight kicks from one match, which is too few to estimate accuracy precisely. The learned landing head improves the count substantially, but on all 39 events, including those without usable input, it recovers only about 60 percent of the supplied landings, so step counts should still be interpreted with care. Finally, the dataset has only three left-footed kickers, so any statement about left-foot performance rests on two or three cases.

9.2 Assessment against the principles of McGowan and colleagues

McGowan et al. [18] describe six principles for judging the quality of a data analysis. On data matching and quality, we audited the inherited labels and withdrew results that relied on unverified AI-generated labels, and every output is bound to a hashed source file. On exhaustiveness, we tested several alternatives for each weak module, including two detectors, mirrored pose, a higher-resolution source, SAM 2, learned heads and augmentation, and reported all of them. On scepticism, we compared every model with a simple control such as the always-right foot or the training majority, chose the foot model by macro-F1 rather than raw accuracy, and kept a frozen test set untouched until the end. On second-order questions, we asked why features were missing rather than only how often, which led to the distinction between model errors and unobservable features in RQ1. On clarity, every unknown carries a recorded reason, and coverage is always reported separately from accuracy. On reproducibility, the saved results can be regenerated, and the portable package was verified on a separate machine. Our main weakness against these principles is the single human reviewer, which limits how sceptical we can be of our own reference.

9.3 Conclusion

On the development footage, the pipeline extracts contact time, ball trajectory and outcome for most kicks, with partial coverage of kicking foot, goalkeeper direction and approach steps. Temporal rules and paired pose checks improve the rule-based outputs; learning small models from human labels produces the largest gains. Data augmentation provides no further improvement under the controlled comparisons. Strict dive height, final ball position and camera-angle timing remain unresolved. The independent test shows that end-to-end execution and strong development agreement do not establish reliable performance on a new match. Further progress requires more diverse footage and independently reviewed labels, particularly for landings, goalkeeper height, outcomes on short clips and the four requested camera categories.

Appendix A Camera angle timing requirement

The brief requires a separate script that accepts a penalty event and returns the time intervals for four specified camera categories. This requirement is distinct from camera-motion stabilisation and pitch-plane projection in Section 7.6.

Table A1 Camera categories and current delivery evidence

Required camera category	Current evidence and status
From behind the kicker	Unresolved. No category-specific start and end times are presented.
From 45 degrees on the kicker's right	Unresolved. No category-specific start and end times are presented.

Required camera category	Current evidence and status
Slow-motion close-up of the kicker	Unresolved. Slow-motion classification has no confirmed outputs in Table 2.
Slow motion from behind the goalkeeper	Unresolved. Slow-motion classification has no confirmed outputs in Table 2.

Table 2 records zero slow-motion classification outputs across 320 camera intervals because cross-shot alignment was not established. Section 9.1 also identifies unresolved slow-motion and camera-type labels. These interval counts therefore do not establish that the four requested categories have been identified.

To close this requirement, the deliverable needs an identified script entry point, a documented invocation, a clear time reference, and reviewed example outputs containing the event identifier, camera category, start time and end time. An absent view should be distinguished from an uncertain classification. No example timings are supplied here because they have not been verified.

References

[1] Chiappori, P.-A., Levitt, S., & Groseclose, T. (2002). Testing mixed-strategy equilibria when players are heterogeneous: The case of penalty kicks in soccer. *American Economic Review, 92*(4), 1138–1151.

[2] Palacios-Huerta, I. (2003). Professionals play minimax. *Review of Economic Studies, 70*(2), 395–415.

[3] Bar-Eli, M., Azar, O. H., Ritov, I., Keidar-Levin, Y., & Schein, G. (2007). Action bias among elite soccer goalkeepers: The case of penalty kicks. *Journal of Economic Psychology, 28*(5), 606–621.

[4] Giancola, S., Amine, M., Dghaily, T., & Ghanem, B. (2018). SoccerNet: A scalable dataset for action spotting in soccer videos. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*.

[5] Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You only look once: Unified, real-time object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 779–788).

[6] Jocher, G., & Qiu, J. (2024). *Ultralytics YOLO11* [Computer software]. https://github.com/ultralytics/ultralytics

[7] Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., & Zitnick, C. L. (2014). Microsoft COCO: Common objects in context. In *European Conference on Computer Vision* (pp. 740–755).

[8] Bazarevsky, V., Grishchenko, I., Raveendran, K., Zhu, T., Zhang, F., & Grundmann, M. (2020). BlazePose: On-device real-time body pose tracking. *arXiv:2006.10204*.

[9] Lugaresi, C., Tang, J., Nash, H., McClanahan, C., Uboweja, E., Hays, M., Zhang, F., Chang, C.-L., Yong, M. G., Lee, J., Chang, W.-T., Hua, W., Georg, M., & Grundmann, M. (2019). MediaPipe: A framework for building perception pipelines. *arXiv:1906.08172*.

[10] Ravi, N., Gabeur, V., Hu, Y.-T., Hu, R., Ryali, C., Ma, T., et al. (2024). SAM 2: Segment anything in images and videos. *arXiv:2408.00714*.

[11] Shi, J., & Tomasi, C. (1994). Good features to track. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 593–600).

[12] Lucas, B. D., & Kanade, T. (1981). An iterative image registration technique with an application to stereo vision. In *Proceedings of the 7th International Joint Conference on Artificial Intelligence* (pp. 674–679).

[13] Fischler, M. A., & Bolles, R. C. (1981). Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM, 24*(6), 381–395.

[14] Duda, R. O., & Hart, P. E. (1972). Use of the Hough transformation to detect lines and curves in pictures. *Communications of the ACM, 15*(1), 11–15.

[15] Hartley, R., & Zisserman, A. (2004). *Multiple View Geometry in Computer Vision* (2nd ed.). Cambridge University Press.

[16] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 770–778).

[17] Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics, 29*(5), 1189–1232.

[18] McGowan, L. D., Peng, R. D., & Hicks, S. C. (2023). Design principles for data analysis. *Journal of Computational and Graphical Statistics, 32*(2), 754–761.

Declaration of AI Use

AI assistants were used to help inspect existing results, write and test parts of the code, and draft and edit the report and presentation materials. The AI-generated step labels identified during the audit in Section 5.1 were withdrawn and are not used as reference labels in this report. The group remains responsible for the submitted code, text, numerical claims and references, including the final review of AI-assisted edits.